THE FIRST COHOMOLOGY OF THE MULTIPLICATIVE GROUP IS THE PICARD GROUP FOR ALL THE BIG TOPOLOGIES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed. The author writes H_etale^1 with the chapter's own \etale macro and the bare subscript 'syntomic'; both are reproduced verbatim.

Editorial note. The preamble is the chapter's own native preamble excerpt used for this source: the Xy-pic setup at lines 14-18 and the theorem declarations with the mathematical macros at lines 154-217 of sources/stacks-etale-cohomology/preamble.tex. The quoted class line and the xr-hyper externaldocument block are not included because the assembler supplies a plain amsart wrapper; no macro was invented or redefined.

Editorial note. The same-chapter pointer is located at its \label line inside the pinned chapter file; the three cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The big and small sites of a scheme and their structure sheaves. Same chapter, Section labelled section-big-small (Stacks tag 03X7); the proof lets tau run over the topologies considered there. Supplies the sites (Sch/X)_tau and their structure sheaves O_tau for tau in {fppf, syntomic, smooth, etale, Zariski}; the chain of identifications is the counted content.
- The structure sheaf of a site computes the given sheaf of functions and Picard groups of the topos. Cohomology on Sites, Lemma labelled sites-cohomology-lemma-h1-invertible (canonical Stacks tag 040E); invoked at proof lines 2513-2515. Supplies H^1_tau(X, G_m) = H^1_tau(X, O_tau^*) = Pic(O_tau) for each topology tau.
- Invertible modules are quasi-coherent, and quasi-coherent modules on the big site do not depend on the topology. Same chapter, Theorem labelled theorem-quasi-coherent (source line 1416, Stacks tag 03OJ) and Descent, Proposition labelled descent-proposition-equivalence-quasi-coherent (Stacks tag 03DX); invoked at proof lines 2522-2524. Supply that an invertible O_tau-module is quasi-coherent and that Pic(O_tau) = Pic(X); the organisation of the chain is retained in the counted span.

Source reference. Original section 03X7: sources/stacks-chow/tags.txt, source lines 5138–5138. Cross-chapter pointer to the section of the same chapter that fixes the list of topologies over which tau runs at proof line 2512; the uniform argument for all topologies is retained verbatim in the counted span.

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Source reference. Original lemma 040E: sources/stacks-chow/tags.txt, source lines 5253–5253. Cross-chapter standard prerequisite invoked at proof lines 2513-2515 for H^1_tau(X, G_m) = H^1_tau(X, O_tau^*) = Pic(O_tau); the uniformity over tau and the subsequent comparison with Pic(X) are retained in the counted span.

Source reference. Original theorem 03OJ: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 1416–1416. Same-chapter pointer offered at proof lines 2523-2524 as one of two ways to see that Pic(O_tau) = Pic(X); the other route (the descent proposition) is cited in the same sentence and the organisation of the chain is retained in the counted span.

Source reference. Original proposition 03DX: sources/stacks-chow/tags.txt, source lines 4436–4436. Cross-chapter standard prerequisite invoked at proof lines 2523-2525 as the more precise statement that quasi-coherent modules on the big site are the same for every topology; it supplies that comparison only.

Theorem 0.1. *For any scheme X we have canonical identifications*

$$\begin{aligned} H^1_{fppf}(X, \mathbf{G}_m) &= H^1_{syntomic}(X, \mathbf{G}_m) \\ &= H^1_{smooth}(X, \mathbf{G}_m) \\ &= H^1_{étale}(X, \mathbf{G}_m) \\ &= H^1_{Zar}(X, \mathbf{G}_m) \\ &= \mathrm{Pic}(X) \\ &= H^1(X, \mathcal{O}_X^*) \end{aligned}$$

Proof. Let τ be one of the topologies considered in Section 03X7. By Cohomology on Sites, Lemma 040E we see that $H^1_\tau(X, \mathbf{G}_m) = H^1_\tau(X, \mathcal{O}_\tau^*) = \mathrm{Pic}(\mathcal{O}_\tau)$ where $\mathcal{O}_\tau$ is the structure sheaf of the site $(Sch/X)_\tau$. Now an invertible $\mathcal{O}_\tau$-module is a quasi-coherent $\mathcal{O}_\tau$-module. By Theorem 03OJ or the more precise Descent, Proposition 03DX we see that $\mathrm{Pic}(\mathcal{O}_\tau) = \mathrm{Pic}(X)$. The last equality is proved in the same way. □